

Topic 3 LMS Assessment

Deep Dive into the n8n AI Agent Node & Core Components

Project: Multi-Component Customer Assistant

Platform: n8n

Assessment Focus: AI Agent configuration, memory recall, calculator tool integration, structured output, and workflow validation.

1. Assessment Overview

This project implements a multi-component AI Agent in n8n using a Chat Trigger, AI Agent, OpenAI Chat Model, Window Buffer Memory, Calculator Tool, and Structured Output Parser.

2. Workflow Configuration

Component	Configuration / Purpose
Chat Trigger	Receives customer messages and starts the workflow.
AI Agent	Customer Assistant Agent with system instructions and maxIterations set to 5.
OpenAI Chat Model	GPT-4o-mini; temperature 0.0; max tokens 1000.
Window Buffer Memory	Maintains conversation context with a context window length of 10.
Calculator Tool	Used for arithmetic such as totals and discounts.
Structured Output Parser	Requires response, category, calculation_used, and result fields.

3. System Prompt

The Customer Assistant is instructed to maintain conversation context, use the calculator tool whenever arithmetic is required, always return the required structured JSON format, remain concise and professional, and set calculation_used according to whether the calculator tool was invoked.

4. Initial Workflow Validation

An initial calculation test was successfully executed. The test used 3 items at \$19.99 each with a 10% discount. The workflow returned a sales category, calculation_used=true, a result of 53.973, and a customer-facing rounded total of \$53.97. This demonstrates successful calculator integration and structured output.

5. Memory Validation Method

Memory recall must be demonstrated through the live Chat Trigger chat interface using one continuous conversation. Separate one-shot workflow executions can receive different session IDs and therefore do not provide reliable evidence of cross-turn memory recall. The live chat demonstration should use consecutive turns in the same session.

6. Recommended Loom Demonstration

Step	Demonstration
1	Open the active/published n8n workflow and briefly show the complete node configuration.
2	Open the Chat Trigger live chat interface.
3	Send: "Hi, my name is Alex."
4	Send: "What is my name?" and show that Alex is recalled.
5	Send: "I want to buy 4 items at \$29.99 each. Apply a 20% discount. What is my final total?"
6	Show the calculator/tool execution and structured JSON result.
7	Send: "What was my name and what total did you calculate?" to demonstrate multi-turn context.
8	Show the workflow execution and configuration for Max Iterations = 5.

7. Evidence / Screenshot Checklist

- Complete workflow showing all connected components.
- Chat Trigger configuration.
- AI Agent configuration showing Max Iterations = 5.
- OpenAI Chat Model showing Temperature = 0.0 and Max Tokens = 1000.
- Window Buffer Memory connection.
- Calculator Tool connection and execution evidence.
- Structured Output Parser schema.
- Live multi-turn chat showing memory recall.
- Structured JSON response.
- Successful workflow execution.

8. Final Submission Deliverables

The assessment requires a mandatory Loom/YouTube demo, a public GitHub repository, an exported n8n workflow JSON, mandatory workflow screenshots, and a README. Documentation in PDF/DOCX is recommended.

9. Assessment Status

The workflow configuration is in place and the initial calculator/structured-output execution has been successful. Before LMS submission, the live continuous-chat memory test, calculator demonstration, structured JSON evidence, and Max Iterations validation should be captured in the Loom recording and screenshots.

10. Reference

Prepared directly from the Topic 3 LMS Assessment requirements and the provided n8n workflow configuration.